

Sampling Stability of Investor-Type Behavioral Coefficients in Aggregate Trading Flows

Symmetric Preference Recovery with Four-Stage Validation

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CCS CONCEPTS

• **Applied computing** → **Economics**; • **Theory of computation** → *Machine learning theory*.

KEYWORDS

investor-type trading flows, penalized regression, sampling stability, combinatorial purged cross-validation, behavioral finance

1 METHOD: SYMMETRIC PREFERENCE RECOVERY

We first define investor behavior and the one-day-ahead forecasting task, and then construct the state features and market context used to explain that behavior. We apply the same conditional linear model to all three investor types and evaluate the sampling stability of its coefficients through four validation procedures. We refer to this framework as *Symmetric Preference Recovery* (SPR). The recovered coefficients represent observed conditional associations rather than structural preferences.

1.1 Behavior Forecasting Problem

Let $\mathcal{I} = \{\text{foreign, institutional, individual}\}$ denote the set of investor types. For investor type $i \in \mathcal{I}$ on trading day t , let $V_{i,t}^{buy}$ and $V_{i,t}^{sell}$ be the total purchase and sale values. We define normalized net-buying behavior as

$$u_{i,t} = \frac{V_{i,t}^{buy} - V_{i,t}^{sell}}{V_{i,t}^{buy} + V_{i,t}^{sell}} \in [-1, 1] \quad (1)$$

This ratio scales net purchases by total trading value and preserves both direction and intensity. The prediction target is $a_{i,t+1} = u_{i,t+1}$. The input state $s_{i,t} = (x_{i,t}, C_t)$ contains only information observed through day t .

All investor types share the same feature vector $x_{i,t} \in \mathbb{R}^3$, market context $C_t \in \mathbb{R}^2$, and model specification; only the parameters are estimated separately. Cross-type differences therefore reflect conditional responses learned along common dimensions, rather than differences imposed by type-specific feature sets.

1.2 State Features and Market Context

The state vector $x_{i,t} = (x_t^{mom}, x_{i,t}^{cross}, x_{i,t}^{uw})^\top$ contains three features.

Medium-term momentum. Let P_t be the closing price of Samsung Electronics. We use the 20-day log return

$$x_t^{mom} = \log(P_t/P_{t-20})$$

with a prespecified horizon. A positive (negative) coefficient is consistent with trend-following (contrarian) behavior [1, 3].

Lagged cross-investor association. For the two types $j, k \neq i$, define $x_{i,t}^{cross} = (u_{j,t-1} + u_{k,t-1})/2$. Using lagged behavior excludes the mechanical offset among investor flows on the same trading day. This feature captures a lagged cross-type relationship and is not a direct test of within-institution herding [4].

Underwater position relative to average cost. Let Q^b and Q^s denote total purchase and sale quantities, and let M^b and M^s denote their corresponding values. The purchase VWAP is $P^b = M^b/Q^b$. In the following recursion, we suppress the common indices (i, t) and write $[v]_+ = \max(v, 0)$:

$$\begin{aligned} \tilde{H} &= \rho H_{t-1}, & H &= [\tilde{H} + Q^b - Q^s]_+, \\ O &= [\tilde{H} - Q^s]_+, & N &= H - O, \\ \bar{P} &= \frac{O\bar{P}_{t-1} + NP^b}{H}, & P^{vwap} &= \frac{M^b + M^s}{Q^b + Q^s}. \end{aligned} \quad (2)$$

Here, $\tilde{H}$ is the decayed inventory carried into day t ; O is the portion of old inventory remaining after sales; N is newly retained inventory; and $H = O + N$ is total effective inventory. The updated average cost is $\bar{P}$, while P^{vwap} is the same-day transaction VWAP. We initialize $H_{i,0} = 0$ and omit zero-quantity terms from the average-cost calculation. We define the underwater feature as

$$x_{i,t}^{uw} = \left[(\bar{P}_{i,t} - P_{i,t}^{vwap}) / \bar{P}_{i,t} \right]_+.$$

It is set to zero when effective inventory is zero or no trade occurs. The forgetting factor $\rho = 0.98$ gives recent trades greater effective weight and implies a half-life of approximately 34 trading days. This aggregate proxy may be sign-consistent with the disposition effect, but it is not an account-level cost basis [5, 6].

Market context. Let r_t be the one-day simple return on the KOSPI 200 and e_t the closing KRW/USD exchange rate. Define $z_t^{FX} = (\log e_t - \mu_{t-1,252}) / \sigma_{t-1,252}$, where $\mu_{t-1,252}$ and $\sigma_{t-1,252}$ are the population mean and standard deviation of $\{\log e_s : s = t - 252, \dots, t - 1\}$. The ordered context vector is $C_t = (r_t, z_t^{FX})^\top$. All features and contexts are standardized using only the training indices of each split. Feature-context correlations range from -0.29 to 0.22 ; we therefore interpret each coefficient as a conditional association holding the other terms fixed.

1.3 Conditional Behavior Model

For investor type i , the score describing next-day behavior is

$$q_{i,t} = (\beta_i + B_i C_t)^\top x_{i,t} + \alpha_i^\top C_t \quad (3)$$

where $\beta_i \in \mathbb{R}^3$ measures feature sensitivity in the average market context, $B_i \in \mathbb{R}^{3 \times 2}$ captures context-dependent changes in that sensitivity, and $\alpha_i \in \mathbb{R}^2$ captures the direct effect of the context when the features are at their standardized means. Thus, $\beta_i + B_i C_t$ is the effective feature-weight vector at time t .

Let rvec denote row-wise vectorization and define

$$w_{i,t} = [x_{i,t}; \text{rvec}(x_{i,t} C_t^\top); C_t], \quad \theta_i = [\beta_i; \text{rvec}(B_i); \alpha_i]$$

so that $q_{i,t} = \theta_i^\top w_{i,t}$ and $\theta_i \in \mathbb{R}^{11}$. The predicted behavior is $\hat{a}_{i,t+1} = \text{clip}(\hat{\theta}_i^\top w_{i,t}, -1, 1)$.

1.4 Estimation and Stability Validation

Let $\mathcal{D}_i$ denote the training indices for investor type i . We estimate the coefficients by L_1 -penalized least squares:

$$\hat{\theta}_i = \arg \min_{\theta} \left\{ \frac{1}{|\mathcal{D}_i|} \sum_{t \in \mathcal{D}_i} (w_{i,t}^\top \theta - a_{i,t+1})^2 + \lambda_i \|\theta\|_1 \right\} \quad (4)$$

$\lambda_i \geq 0$ is the type-specific penalty strength. The objective in Eq. (4) is convex, and coordinate descent obtains a global solution. To verify uniqueness, define $n_i = |\mathcal{D}_i|$, $W_i = [w_{i,t}^\top]_{t \in \mathcal{D}_i}$, $y_i = [a_{i,t+1}]_{t \in \mathcal{D}_i}$. For $\lambda_i = 0$, we verify that $\text{rank}(W_i) = 11$. For $\lambda_i > 0$, define the KKT vector $g_i = 2W_i^\top (y_i - W_i \hat{\theta}_i)/n_i$ and set $E_i = \{j : |g_{i,j}| = \lambda_i\}$. We then verify $\text{rank}(W_{i,E_i}) = |E_i|$. The solution is unique for all three investor types in all 45 splits, with a worst-case condition number of 6.86.

Rather than interpreting a single fitted coefficient vector, we evaluate sampling stability through four procedures.

(V1) Combinatorial purged cross-validation. We construct 45 CPCV splits [2] by selecting two of ten chronological folds for testing. We apply a one-day purge and a five-day embargo, and fit every standardization step using training indices only. Because the splits share observations, sign consistency is treated as a stability measure rather than a p -value.

(V2) Calendar-month block bootstrap. To address this dependence, we resample calendar-month blocks with replacement 1,000 times and construct 95% confidence intervals. Monthly blocks preserve within-block serial dependence in daily investor flows.

(V3) Expanding walk-forward evaluation. Because a CPCV test fold may precede some training folds, we also conduct ten expanding window evaluations. Starting with 400 observations, each window applies a five-day purge, predicts the next 60 trading days, and then expands the training sample by 60 observations.

(V4) Penalty substitution. To verify that the results are not artifacts of the L_1 penalty, we re-estimate the model using an L_2 penalty and no penalty, and compare coefficient signs across specifications.

The evaluation metrics are directional accuracy, the correlation between predicted and realized behavior, MAE, and out-of-sample R^2 . Directional accuracy is the proportion of observations for which the predicted and realized signs agree; we set $\text{sign}(0) = 0$. The baseline prediction for out-of-sample R^2 is the mean behavior in the training sample of each split. Because the dispersion of behavior differs across investor types, absolute MAE values are not used to rank the types.

Table 1: CPCV out-of-sample performance (mean $\pm$ standard deviation across 45 splits). The table checks that the model does not learn noise alone; it does not establish predictive superiority.

Metric	Foreign	Institutional	Individual
Directional accuracy	0.636 $\pm$ 0.040	0.536 $\pm$ 0.035	0.602 $\pm$ 0.044
Correlation	0.279 $\pm$ 0.106	0.060 $\pm$ 0.062	0.239 $\pm$ 0.116
MAE	0.195	0.095	0.266
Out-of-sample R^2	0.105	0.002	0.102

2 EXPERIMENTS

2.1 Data

We use the latest 973 daily observations for Samsung Electronics (005930) for which the 252-day exchange-rate history, 20-day momentum, and next-day behavior can all be computed. The state dates span January 6, 2022, to December 29, 2025. The single-stock design provides a controlled setting that removes cross-stock fixed effects and liquidity differences, thereby isolating heterogeneity across investor types.

Mean behavior is close to zero for all three types, but its standard deviation is 0.334 for individual investors, 0.257 for foreign investors, and 0.125 for institutions. Their first-order autocorrelations are 0.345, 0.401, and 0.149, respectively. Same-day behavior is strongly negatively correlated across types (foreign-individual: -0.772 ; institutional-individual: -0.499). The gap between the sum of net purchases by the three types and the market total, which includes other corporations and other foreign entities, averages KRW 14.4 billion per day, or 1.3% of trading value. The type-specific values of λ are 0, 0.01, and 0.0003. They were carried over from a preceding specification rather than selected by nested cross-validation in the present specification. Assigning 0, 0.0005, or 0.01 as a common penalty for all three types does not alter the sign classifications reported below.

2.2 Coefficient and Sampling Stability (V1–V4)

The dispersion across splits in Table 1 precludes comparisons based on any single split. In Figure 2, predictions for foreign and individual investors follow broad trading-flow regimes, whereas institutional predictions remain near zero. Coefficient stability shows the same contrast (Figure 1).

(V1) The foreign momentum weight is +0.0502 and positive in all 45 splits, whereas the individual weight is -0.0383 and negative in every split. Lagged cross-flow coefficients are negative for all three types in every split. The individual underwater coefficient is positive in 97.8% of splits. For institutions, the Lasso sets momentum and underwater coefficients exactly to zero in 36 and 44 of the 45 splits, respectively.

(V2) Eight coefficients have bootstrap intervals that exclude zero: momentum and lagged cross-flow, together with both direct context effects, for foreign and individual investors. The individual underwater interval ranges from -0.0122 to $+0.0665$, and all institutional intervals include zero. Despite complete sign consistency across the 45 CPCV splits, the individual underwater coefficient

Sampling Stability of Coefficients by Investor Type

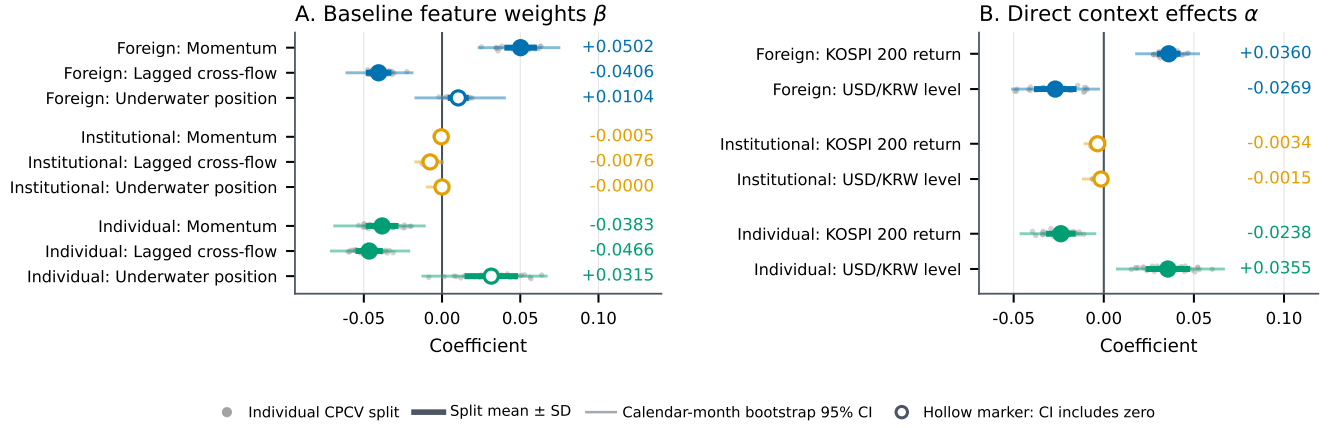


Figure 1: Sampling stability of the recovered coefficients. Gray dots denote the 45 CPCV splits, thick lines show the standard deviation across splits, and thin lines show calendar-month block-bootstrap 95% confidence intervals. Hollow markers indicate intervals containing zero; values on the right are mean coefficients across splits.

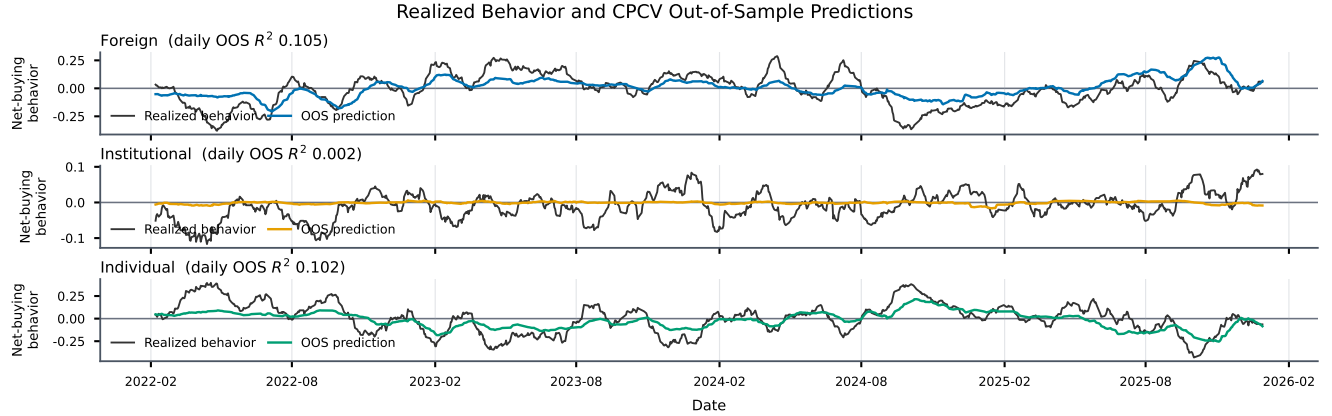


Figure 2: Realized behavior and CPCV out-of-sample predictions by investor type. We first average the nine CPCV test predictions available for each date and then apply a 20-trading-day moving average to both realized and predicted behavior. Panel-level out-of-sample R^2 values are the 45-split means for unsmoothed daily predictions and match Table 1.

changes sign in 5.5% of calendar-month resamples. This result illustrates the warning in Section 1.4: because CPCV splits share observations, sign consistency alone can understate uncertainty.

(V3) Expanding walk-forward performance varies considerably across windows. Foreign directional accuracy is 0.610 ± 0.116 , and its correlation is 0.131 ± 0.162 , less than half the CPCV correlation. We report this decline as a temporal generalization limit. The coefficient paths are more stable. Foreign momentum remains positive and individual momentum remains negative in all ten windows; the two paths never cross. The same contrast holds for both direct context effects, while the institutional paths remain near zero.

(V4) Foreign and individual coefficients agree to four decimal places under L_1 , L_2 , and unpenalized estimation. Only institutional

coefficients move with the penalty; for example, lagged cross-flow changes from -0.0076 to -0.0124 .

Robustness. Across the nine combinations of forgetting factors $\rho \in \{0.95, 0.98, 0.99\}$ and momentum windows $\{10, 20, 60\}$, foreign and individual momentum coefficients are positive and negative, respectively, in 100% of specifications. The individual underwater coefficient is positive in at least 91%. Imposing a common λ across investor types leaves these sign classifications unchanged.

3 INTERPRETATION

- **Foreign investors: stable conditional associations with market direction and the exchange-rate regime.** The momentum weight of +0.0502 is positive in all 45

CPCV splits, and its calendar-month bootstrap interval excludes zero. The association between recent price increases and greater foreign net buying is therefore insensitive to the split and resampling scheme. The KOSPI 200 effect of +0.0360 and the exchange-rate effect of -0.0269 pass the same checks. Conditional on the other features, foreign investors tend to buy on market up-days and sell when the KRW is weak relative to the preceding 252 days. These signs accord with prior evidence on foreign trend trading [1, 3]. The design does not separate counterparty flows from common market shocks, so these patterns do not identify an independent structural preference. For the same reason, the stable negative lagged cross-flow coefficient is not direct evidence of anti-herding or imitation across investor types.

- **Individual investors: opposing market associations and limited loss avoidance.** Individual momentum (-0.0383), the KOSPI 200 effect (-0.0238), and the exchange-rate effect ($+0.0355$) have signs opposite to their foreign counterparts. They remain stable across CPCV splits, calendar-month resampling, walk-forward evaluation, and penalty substitution. Individual investors thus tend to sell after stock and market increases and to buy during KRW-weakness regimes. The strong negative same-day correlation between foreign and individual flows prevents this contrast from identifying an independent individual contrarian preference. The underwater weight is $+0.0315$ and positive in 97.8% of CPCV splits and at least 91% of robustness specifications. A deeper loss relative to average cost is therefore associated with greater net buying or reduced selling, a sign consistent with avoiding loss realization or averaging down. However, the calendar-month interval includes zero, and aggregate flows cannot distinguish additional purchases from deferred sales. We interpret the coefficient as a limited, repeatedly observed signal rather than direct evidence of loss aversion.
- **Institutions: no common behavioral weight in aggregate flows.** The data do not reveal which information individual institutions use. Within the aggregate Korean “institutional” category, however, the specified features do not form a stable common trading rule. The instability of all main coefficients and an out-of-sample R^2 of 0.002 are consistent with heterogeneous institutional responses offsetting one another during aggregation.

Synthesis. Foreign and individual momentum, market, and exchange rate coefficients point in opposite directions and remain stable across sample constructions. Individual underwater positions retain a limited signal consistent with avoidance of loss realization, whereas aggregate institutional flows reveal no common behavioral rule. The central result is that these differences recur under a symmetric estimation design and four stability checks. Decomposing their origins into independent preferences, counterparty absorption, and common shocks lies outside the identification scope of the analysis.

4 LIMITATIONS AND FUTURE RESEARCH

(1) Interpretive scope. The recovered weights are conditional associations rather than causal effects. The model does not include returns, Sharpe ratios, or transaction costs; Table 1 therefore does not establish the superiority of a trading rule or flow forecast. Because the design does not separately identify the accounting offset in same-day flows and common market shocks, the opposing foreign and individual coefficients and the negative lagged cross-flow coefficients cannot be interpreted as independent preferences.

(2) Disposition-effect proxy. The individual underwater coefficient does not pass the calendar-month block bootstrap. Moreover, the average cost is an approximation based on aggregate trades rather than realized gains and losses at the account level. Its sign is consistent with the disposition effect, but it does not constitute a direct test.

(3) Generalizability. The single-stock setting controls for cross-stock heterogeneity but has not been validated across other securities or markets.

Future research. First, applying the same estimator and V1–V4 procedures to stocks with different foreign ownership and liquidity would test cross-sectional generalizability. Second, distinguishing the accounting offset in simultaneous flows from each investor type’s direct response requires richer data or an alternative identification design. Third, the hypothesis that aggregation produces institutional instability can be tested directly by applying symmetric estimation to the KRX’s disaggregated investor categories.

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